

## A highly informative microsatellite panel for individual identification and sex determination of jungle cats (*Felis chaus*)

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**Abstract** The jungle cat (*Felis chaus*), a small-sized felid, is distributed across Southeast and South Asia to Egypt in Africa. In India, it is one of the most common small cat species but no reliable methods are available to monitor its population status. We describe a non-invasive genetic monitoring technique using fecal samples for individual identification and sex determination of jungle cats. We evaluated 21 feline microsatellites and optimized a panel of 11 highly polymorphic microsatellite loci that yield a cumulative Probability of Identity between siblings value of  $5.51 \times 10^{-6}$ . We used this panel to identify 110 individuals from 118 jungle cat scats collected from tiger reserves in Central India. We identified 22 males and 35 females by amplifying a fragment of the Amelogenin protein gene. This panel will be helpful to study genetic structure, gene flow, relatedness, sex ratio, and population estimation in jungle cats.

**Keywords** Noninvasive DNA sampling · *Felis chaus* · Individual identification · Microsatellite · Sex assignment

Non-invasive genetic sampling holds promise for monitoring rare and elusive species (Waits 2004). Here we describe a non-invasive method using fecal samples and microsatellite markers (also known as variable Numbers of tandem repeats—VNTRs) to identify individuals and determine sex in jungle cats.

During April–June 2009 and Nov 2009–May 2010, we collected fecal samples of wild jungle cats in five Tiger Reserves (TR) in Central India: Kanha (KTR), Satpura (STR) (in Madhya Pradesh); Melghat (MTR) (in Maharashtra); Pench (PTR, in both the states); and corridors connecting these reserves. These TRs are separated by distances ranging from 115 to 370 km. We sampled along forest roads and trails and collected all small carnivore scats of different ages (fresh as well as old), identified by their morphology (size, shape, and diameter). The sampling methods were designed to maximize the number of individuals encountered and not to recapture individuals. We stored these samples in 100 % ethanol until DNA isolation in the lab.

We isolated genomic DNA using the QIAamp mini-stool kit (Qiagen Inc.) following the manufacturer's protocol. We targeted portions of the outermost layer of the scat sample for DNA extractions because it contains high proportion of epithelial cells from the intestinal lining. In the wild, jungle cat scat could be confused with that of the golden jackal (*Canis aureus*) based on morphometry (Vanak and Mukherjee 2008). Also, jungle cats often coexist with domestic cats (*Felis silvestris catus*) in most of their range outside and along the periphery of the protected areas. To confirm the feline origin of the scats, we first screened DNA samples by using fluorescently-labeled felid-specific 16S rRNA primers, which amplify a 210 bp fragment (Mukherjee et al. 2010). To identify and differentiate jungle cat scats from those of domestic cats, we used the PCR–RFLP technique (Mukherjee et al. 2010) in

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**Table 1** Characteristics of 21 microsatellites screened on 8 individual jungle cats

| Locus               | Repeat type | Ta | Size range | Success | No. of alleles | Ho   | He   | PID       | PID (SIBS) | PID (cum) | PID (SIBS)cum | ADO   | FA    | Chromosomal assignment |
|---------------------|-------------|----|------------|---------|----------------|------|------|-----------|------------|-----------|---------------|-------|-------|------------------------|
| FCA008 <sup>a</sup> | Di          | 55 | 126–132    | 0.78    | 3              | 0.44 | 0.61 | 1.83E–01  | 5.02E–01   | 1.83E–01  | 5.02E–01      | 0.063 | 0.05  | A1                     |
| F041 <sup>a</sup>   | Tetra       | 55 | 107–147    | 1       | 7              | 0.89 | 0.73 | 2.04E–02  | 4.12E–01   | 3.74E–03  | 2.07E–01      | 0.063 | 0     | D2                     |
| F042 <sup>a</sup>   | Tetra       | 55 | 211–251    | 1       | 7              | 0.56 | 0.8  | 1.18E–02  | 3.69E–01   | 4.41E–05  | 7.64E–02      | 0.1   | 0     | A1                     |
| F043 <sup>a</sup>   | Di          | 55 | 106–118    | 1       | 5              | 0.67 | 0.75 | 5.11E–02  | 4.03E–01   | 2.25E–06  | 3.08E–02      | 0.042 | 0     | C2                     |
| FCA077 <sup>a</sup> | Di          | 55 | 128–152    | 1       | 8              | 0.78 | 0.81 | 7.32E–03  | 3.61E–01   | 1.65E–08  | 1.11E–02      | 0.071 | 0     | C2                     |
| FCA090 <sup>a</sup> | Di          | 55 | 99–119     | 0.89    | 6              | 0.56 | 0.75 | 3.55E–02  | 4.00E–01   | 5.84E–10  | 4.45E–03      | 0.05  | 0     | A1                     |
| FCA126 <sup>a</sup> | Di          | 55 | 116–146    | 1       | 7              | 0.78 | 0.83 | 6.20E–03  | 3.49E–01   | 3.62E–12  | 1.55E–03      | 0.036 | 0     | B1                     |
| FCA279 <sup>a</sup> | Di          | 55 | 93–109     | 1       | 7              | 0.56 | 0.78 | 2.31E–02  | 3.82E–01   | 8.35E–14  | 5.93E–04      | 0.05  | 0     | C1                     |
| FCA441 <sup>a</sup> | Tetra       | 55 | 116–146    | 1       | 5              | 0.67 | 0.67 | 6.62E–02  | 4.56E–01   | 5.53E–15  | 2.70E–04      | 0.042 | 0     | D3                     |
| FCA628 <sup>a</sup> | Di          | 56 | 93–127     | 1       | 10             | 1    | 0.88 | –1.61E–03 | 3.15E–01   | –8.91E–18 | 8.51E–05      | 0.028 | 0     | D2/E2                  |
| FCA672 <sup>a</sup> | Di          | 55 | 82–104     | 1       | 9              | 0.78 | 0.85 | –3.10E–03 | 3.37E–01   | 2.76E–20  | 2.87E–05      | 0.036 | 0.125 | F2                     |
| F115                | Tetra       | 55 | 200–264    | 0.22    | 3              | 0.11 | 0.63 | 1.47E–01  | 4.90E–01   | 4.06E–21  | 1.41E–05      | 0.25  | 0     | B2                     |
| FCA146              | Di          | 55 | 104–114    | 0.11    | 2              | 0.11 | 0.5  | 3.67E–01  | 5.94E–01   | 1.49E–21  | 8.35E–06      | 0     | 0     | A1                     |
| FCA170              | Di          | 55 | 102–114    | 0.78    | 4              | 0.44 | 0.73 | 6.41E–02  | 4.12E–01   | 9.54E–23  | 3.44E–06      | 0.188 | 0.05  | F2                     |
| FCA205              | Di          | 55 | 110        | 0.11    | 1              | 0    | 0    | 1.00E+00  | 1.00E+00   | 9.54E–23  | 3.44E–06      | 0     | 0     | B3                     |
| FCA232              | Di          | 55 | 107–111    | 0.44    | 3              | 0.33 | 0.53 | 1.84E–01  | 5.55E–01   | 1.75E–23  | 1.91E–06      | 0.167 | 0     | B4                     |
| FCA304              | Di          | 55 | 127–154    | 0.67    | 8              | 0.67 | 0.85 | 7.67E–04  | 3.37E–01   | 1.35E–26  | 6.43E–07      | 0.083 | 0     | A2                     |
| E7                  | Tetra       | 55 | 147–157    | 0.67    | 4              | 0.67 | 0.72 | 7.25E–02  | 4.21E–01   | 9.76E–28  | 2.71E–07      | 0.208 | 0     | NA                     |
| FCA052              | Di          | 55 |            | 0       | 0              | 0    | 0    | 0.00E+00  | 0.00E00    | 0.00E+00  | 0.00E+00      | 0     | 0     | A2                     |
| FCA391              | Di          | 55 |            | 0       | 0              | 0    | 0    | 0.00E+00  | 0.00E+00   | 0.00E+00  | 0.00E+00      | 0     | 0     | B3                     |
| FCA453              | Tetra       | 55 |            | 0       | 0              | 0    | 0    | 0.00E+00  | 0.00E+00   | 0.00E+00  | 0.00E+00      | 0     | 0     | A1                     |

Ta, annealing temperature; Success, amplification success in percentage; ADO, allelic drop out; FA, false alleles

<sup>a</sup> Loci selected for final panel

**Table 2** Genetic variability of 110 jungle cats\* at 11 microsatellite loci

| Locus               | Size range | Success | No. of alleles | Ho   | He   | PID      | PID (SIBS) | PID (cum) | PID (SIBS)cum | ADO   | FA    |
|---------------------|------------|---------|----------------|------|------|----------|------------|-----------|---------------|-------|-------|
| FCA008 <sup>a</sup> | 124–144    | 0.82    | 11             | 0.61 | 0.8  | 5.74E–02 | 3.67E–01   | 5.74E–02  | 3.67E–01      | 0.047 | 0.008 |
| F043 <sup>b</sup>   | 106–124    | 0.99    | 9              | 0.65 | 0.77 | 8.14E–02 | 3.86E–01   | 4.67E–03  | 1.42E–01      | 0.021 | 0     |
| FCA126 <sup>c</sup> | 116–150    | 0.99    | 17             | 0.81 | 0.9  | 1.43E–02 | 3.02E–01   | 6.69E–05  | 4.29E–02      | 0.018 | 0     |
| FCA279 <sup>c</sup> | 83–113     | 0.99    | 14             | 0.69 | 0.85 | 3.49E–02 | 3.34E–01   | 2.34E–06  | 1.43E–02      | 0.011 | 0     |
| FCA441 <sup>a</sup> | 97–129     | 0.78    | 9              | 0.61 | 0.8  | 6.13E–02 | 3.66E–01   | 1.43E–07  | 5.23E–03      | 0.019 | 0     |
| FCA628 <sup>d</sup> | 89–131     | 0.93    | 20             | 0.78 | 0.92 | 8.87E–03 | 2.91E–01   | 1.27E–09  | 1.52E–03      | 0.01  | 0     |
| FCA672 <sup>b</sup> | 82–110     | 0.9     | 14             | 0.83 | 0.86 | 2.78E–02 | 3.26E–01   | 3.53E–11  | 4.96E–04      | 0.023 | 0     |
| FCA090              | 87–121     | 0.81    | 16             | 0.4  | 0.84 | 4.26E–02 | 3.43E–01   | 1.50E–12  | 1.70E–04      | 0.046 | 0.012 |
| FCA077 <sup>d</sup> | 124–152    | 0.72    | 15             | 0.49 | 0.9  | 1.48E–02 | 3.03E–01   | 2.23E–14  | 5.16E–05      | 0.068 | 0     |
| F041                | 107–159    | 0.64    | 14             | 0.39 | 0.86 | 3.22E–02 | 3.31E–01   | 7.16E–16  | 1.71E–05      | 0.053 | 0     |
| F042 <sup>b</sup>   | 207–251    | 0.64    | 12             | 0.37 | 0.87 | 2.77E–02 | 3.23E–01   | 1.99E–17  | 5.51E–06      | 0.111 | 0     |

Superscript letters indicate optimized multiplexed reactions (<sup>a</sup> FCA008–FCA441, <sup>b</sup> F043–FCA672–F042, <sup>c</sup> FCA126–FCA279, <sup>d</sup> FCA628–FCA077)

\* Individual jungle cats identified at each tiger reserves are as follows (Kanha = 35, Pench = 30, Satpura = 15, Melghat = 24, and Corridor = 6)

which we used restriction enzyme *ASEI*. This enzyme splits the 210 bp fragment of jungle cat DNA into two fragments of size 91 and 110 bp, while domestic cat DNA does not have a restriction site for this enzyme. We used positive controls of jungle cat, domestic cat, and golden jackal to validate this method. To select a panel of microsatellite loci, we used 8 geographically-isolated jungle-cat-positive samples (1 tissue and 7 scats) to screen 21 microsatellites developed on domestic cats (Menotti-Raymond et al. 1999) and tigers (Bhagavatula and Singh 2006). Some of these microsatellite loci were used in our previous studies on tigers and leopards (Dutta et al. 2012a, b; Sharma et al. 2013). Following individual identification, we ascertained sex using Amelogenin-gene primers (Pilgrim et al. 2005), which are heterozygous in males (194, 214) and homozygous in females (214, 214). This primer has been used on various feline species (Pilgrim et al. 2005) and leopards in our previous study (Dutta et al. 2012a).

For microsatellite loci, we conducted all PCRs in 5 µl reactions, containing 2 µl Multiplex PCR buffer mix (Qiagen), 0.2 µm labelled forward and 0.2 µm unlabelled reverse primer, 0.5 µl of 10× BSA, and 1.5 µl DNA template. PCR conditions for microsatellites were: initial denaturation (94 °C for 15 min), 45 cycles of denaturation (94 °C for 30 s), annealing (Ta for 45 s) and extension (72 °C for 45 s), and a final extension (72 °C for 30 min). We followed Pilgrim et al. (2005) for the PCR protocol for sex identification. We ran all PCR products on an ABI-3730xl sequencer, using GeneScan<sup>TM</sup>–500LIZ<sup>®</sup> size standard. We scored alleles using GeneMapper-4.1 (Applied Biosystems).

We carried out all pre-PCR procedures with aerosol barrier tips in a separate room to minimize cross-contamination. We included negative controls in all extractions and PCR

reactions to detect any contamination. To account for error rates, we followed a modified multi-tube approach (Taberlet et al. 1996), and repeated each PCR (twice for species identification, four times for microsatellites and sex identification).

We used MICRO-CHECKER-2.2.3 (van Oosterhout et al. 2004) to detect loci containing errors due to scoring or stuttering, and to large allele dropout. We used GIMLET-1.3.3 (Valière 2002) to obtain measures of genetic variation such as mean number of alleles per locus, observed (Ho) and expected (He) heterozygosities, and the P<sub>ID</sub> (probability of identity) and P<sub>ID(sibs)</sub> value (Waits et al. 2001). We estimated PCR success and genotyping error rates (false alleles-FA, and allelic drop out-ADO) using GIMLET. We conducted tests for deviations from Hardy–Weinberg equilibrium (HWE) and tests for linkage disequilibrium (LD) in GENEPOP-3.4 (Raymond and Rousset 1995; Rousset 2008) with a Bonferroni correction (Rice 1989). We used CERVUS-3.0.3 (Kalinowski et al. 2007) for individual identification.

In the trial dataset (n = 8), we verified the species and identified 8 individuals. We successfully generated data for 18 microsatellite loci, out of which one locus (FCA205) was monomorphic. Three loci (FCA052, FCA391, FCA453) failed to amplify.

We found no evidence of stuttering errors or large allele dropout in any of the 17 polymorphic microsatellite loci. Amplification success across loci ranged from 11 to 100 %. These loci had a mean of 5.8 alleles and mean Ho of 0.59 (Table 1).

We then selected a panel of 11 microsatellites based on amplification success, allele numbers, Ho, and low genotyping error rates. We optimized them into 4 multiplexed reactions and two single reactions (Table 2).

In the field, we collected 190 scat samples, from which we could assign 118 to jungle cats. Out of these 118 samples, 110 amplified for 6 or more loci and were retained for further analysis. We identified 110 individual jungle cats using the selected panel with  $P_{ID(cum)}$  of  $1.99 \times 10^{-17}$  and  $P_{ID(sibs)(cum)}$  of  $5.51 \times 10^{-06}$  (Table 2).

Amplification success was 93.2 %, and wild jungle cats showed high genetic variation with a mean of 13.73 alleles and mean  $H_o$  of 0.6. Six out of 11 loci (FCA279, FCA441, FCA077, FCA090, F041, F042) deviated from HWE when samples from all TRs were analyzed as a single population. When samples were grouped by their sampling locality, 4 loci in KTR (FCA008, FCA090, F042, FCA279), 3 in PTR (FCA090, F041, F042.), 2 in MTR (FCA090, FCA077), and one locus in STR (F041) deviated from HWE. No loci pair out of 275 pairwise comparisons was in significant LD after Bonferroni corrections. We identified 35 females and 22 males (51.8 % success) by using Amelogenin primer on 110 individuals.

The microsatellite panel we describe is highly polymorphic and had sufficiently low  $P_{ID}$  values for reliable identification of individuals. These loci can be efficiently multiplexed into fewer PCR reactions, thus saving time and money in population genetics and wildlife forensic studies.

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